

Glucagon

(Glucagen)

- I. Classification**
 - Hormone
- II. Action(s)**
 - Converts hepatic glycogen to glucose, thus increasing blood glucose concentration
 - Relaxes smooth muscles of the gastrointestinal tract
- III. Indication(s)**
 - Hypoglycemia
 - Calcium channel blocker overdose, beta-blocker overdose
- IV. Contraindications**
 - Known hypersensitivity
 - Adrenal or pancreatic tumors (pheochromocytoma, insulinoma)
- V. Precautions**
 - Patient must be advised to eat a carbohydrate-heavy meal after use
 - Does not work well in malnourished persons (minimal glycogen stores)
- VI. Drug interactions**
 - Beta blockers decrease effectiveness of Glucagon
 - Use with anticoagulants can increase the risk of bleeding
- VII. Adult Dosing and Administration**
 - Hypoglycemia: 1 mg IV / IO / IM, may repeat once after 5 minutes
 - Only by Medical Control consult
 - Calcium channel blocker or Beta-blocker overdose:
 - 3 mg IV / IO, may give additional 2 mg IV / IO after 10 minutes
- VIII. Pediatric Dosing and Administration**
 - Age <5 year (<20 kg): 0.5 mg IV / IO/ IM
 - Age 5 year and older (>20 kg): 1mg IV / IO / IM
 - May repeat once after 5 minutes
 - Calcium Channel Block or Beta-blocker overdose:
 - 50 mcg/kg IV / IO (max dose 3 mg), may repeat after 10 minutes (max total 5 mg)
- IX. Onset**
 - 5 – 20 minutes
- X. Duration**
 - Peaks 30 minutes post injection
 - Designed to increase blood sugar at least 30 mg/dl in < 5 minutes
- XI. Possible Adverse Effects**
 - Headache and nausea
 - Rash / allergic reaction
 - Hyperglycemia
- XII. Note**
 - IO D10W is preferable to glucagon in an emergency setting due to glucagon depleting glycogen stores and making Glucagon to be less effective in repeat doses
 - This leaves the patient with fewer rescue options at home until Glycogen is replenished